

Module 08: Inheritance and access modifiers

Objective: Create an inheritance hierarchy

System requirements: .Net 5.0

Tools: Visual Studio 2019

Duration: ~30min

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1 Creation of inheriting classes

The goal of this lab is to create new classes that are derived from the vehicle class from the previous labs.

- Create a total of three new classes (car, ship, plane) and let them inherit from the vehicle class.
- Each of these classes should get at least one new class-specific property (e.g. number of seats, draught, maximum altitude, fuel, etc.).
- Write the respective constructors considering the new properties and with reference to the constructor of the vehicle class.
- The Info() methods of the individual classes should now also output the new property (use the virtual/override mechanics and a reference back to the Info() of the vehicle class).

2 Object instantiation

Now create one object of each new class in the Main() and execute the Info() methods.

```
The car VW costs 20000€ and has a maximum velocity of 210km/h. It has 5 doors  
The ship Titanic costs 250000000€ and has a maximum velocity of 40km/h. Its fuel is diesel.  
The plane Boeing costs 900000000€ and has a maximum velocity of 500km/h. It has a maximum altitude of 10000m.
```